5.4 Exploring Corresponding Parts of Congruent Triangles are Congruent

Lesson Introduction

Benchmarks	MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures. MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure. MA.912.GR.2.6 Apply rigid transformation to map one figure onto another to justify that the two figures are congruent.	Learning Targets	• I can relate the definition of congruence in terms of rigid motions to corresponding parts of congruent triangles. • I can use corresponding parts of congruent triangles to write congruence statements.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.	N/A

Essential Questions	• How can I name corresponding parts when two triangles are congruent? • How can Corresponding Parts of Congruent Triangles are Congruent (CPCTC) be used to determine which parts of a triangle are congruent?
Lesson Narrative	In this lesson, students will continue their learning of triangle congruence by extending the scenarios to include the use of rigid motions. Students will also learn the concept of CPCTC. They will practice writing congruence statements of corresponding parts from congruent triangles.

Component Summary				
	5.4.1 Warm-Up (~5 min.)	Notice and wonder about geometry art *(SE p. 241)*	5.4.3 Guided Instruction (15-20 min.)	Formal instruction on CPCTC *(SE p. 243)*
	5.4.2 Collaboration (10-15 min.)	Collaboration on corresponding parts and rigid motions of triangles *(SE p. 242)*	5.4.4 Guided Practice (5-10 min.)	Practice opportunity naming corresponding parts of congruent triangles *(SE p. 245)*
	5.4.5 Wrap-Up (~5 min.)	Cool Down naming corresponding parts of congruent triangles and an Error Analysis *(SE p. 246)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: • Definition of Congruence in Terms of Rigid Motions • Corresponding Parts of Congruent Triangles are Congruent (CPCTC) Additional Terms: N/A	• (Optional) Markers • (Optional) Patty paper	• This lesson guides students through a collaborative activity to review rigid motions and corresponding parts. Consider doing this activity on your own prior to the lesson to highlight potential misconceptions that may arise and to better understand where the activity is going.

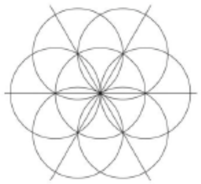
Teacher Tips	**Common Misconceptions** • Students may make mistakes when conducting the rigid motions. • Students may make errors when determining corresponding parts, particularly if the triangles are rotated.	**Things to Consider** • Students should have a working knowledge of rigid motions and how they function. However, if your students struggle with them, consider doing a brief review of the different rigid motions and their properties.
	Literacy and Language Routines **Notice and Wonder** 5.4.1 Warm-Up asks students to review patterns and designs. Students will describe what they notice about a design. Then, students will describe what they wonder about the pattern or final design when they review the last step of the 12-fold pattern.	**Mathematical Thinking and Reasoning (MTR) Standard** **MA.K12.MTR.5.1: Patterns and Structure** Students will connect their prior learning of rigid motions and their properties to determine if two triangles are congruent.
Differentiate Instruction	Consider pairing students based on their performance in the prior set of lessons. Their understanding of what they have learned so far about triangle congruence can be a way to determine which students may need additional support. Consider the use of physical manipulatives or patty paper during 5.4.3 Guided Instruction in order for visual learners to be able to picture which parts correspond with others. 5.4.5 Wrap-Up is a Cool Down that provides students with the opportunity to display mastery of the content from this lesson. Student response data can provide information on the degree to which students have mastered the content of this lesson. Consider having students mark the congruent parts on the figure as part of the data collected to measure mastery.	
Lesson Notes:		

5.4.1 Warm-Up: Art in Geometry

Component Overview: *Students will explore geometry that exists within art and designs. For this activity, students can work independently or with a small group. Consider having students, at a minimum, discuss the things they notice and things they wonder with a partner.*

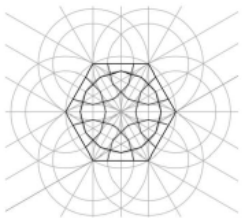
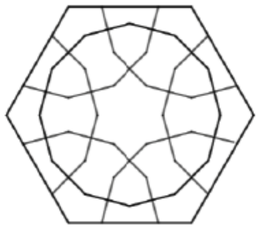
5.4.1 Warm-Up: Art in Geometry

Salih is taking an art class that looks at the different patterns in tiles from across the world. Salih used a compass to create this first step in a 12-fold pattern.



1. What do you notice about the design?
   Answers will vary. I notice that the design is very symmetrical. I also notice that the circles and lines all overlap.

The last step of the 12-fold pattern and the final design are shown.

Last Step	Final Design
	

2. What do you wonder about the pattern or the final design?
   Answers will vary. I wonder how the artist was able to start with circles and end with a hexagon. I also wonder how the pattern repeats so perfectly.

For this activity, consider reading the narrative with the students out loud. It is important, for this context, for students to know that this is a 12-fold pattern prior to answering questions 1 and 2.

Allow students to share out the things that they noticed and wondered. Consider listing things that students share in a chart so that students can hear alternative perspectives.

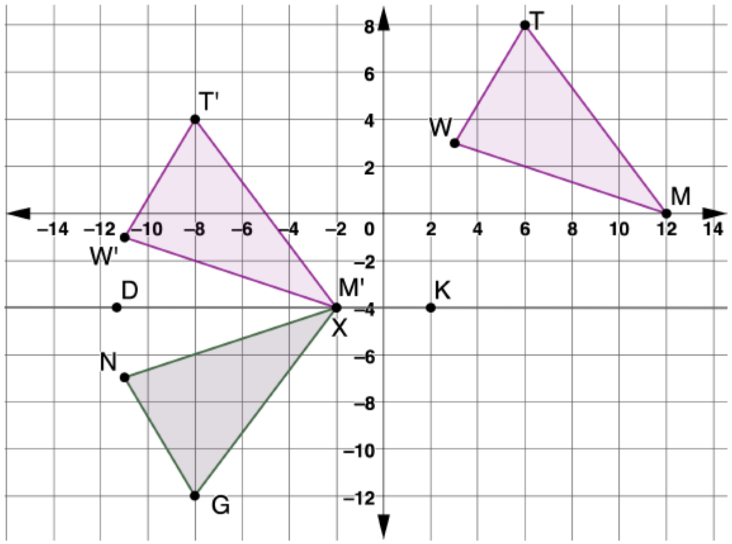
5.4.2 Collaboration: Congruence in Terms of Rigid Motions

Component Overview: *Students will work together to discover corresponding parts and rigid motions of triangles. For this activity, release students to complete questions 1 and 2 with a partner and then come back together as a class to discuss. Review the formal definition for congruence in terms of rigid motion and then release students to continue working with their partners on questions 3 and 4.*

5.4.2 Collaboration: Congruence in Terms of Rigid Motions

Work with your partner on the following.

The diagram shows a sequence of transformations that, when applied to ΔTMW, show that ΔTMW maps to ΔGXN.



1. What are the transformations used to map ΔTMW to ΔGXN?
   The transformation was a slide 14 units to the left and 4 units down. Next the triangle was reflected over the line $\overleftrightarrow{DK}(y = -4)$

If students have struggled with rigid motions in the past, consider doing a brief review of basic rigid motions prior to launching this activity. If you do this, be cautious of your time so as to not make it a full lesson on rigid motions in isolation.

Notice and Wonder
What are some things that you notice about the triangles on the coordinate grid? What are some things that you wonder about them?

Some students may describe the transformations as two separate translations and a reflection instead of one translation and a reflection, as seen in the answer key. Both are correct and should be accepted. The student's response here determines how they will respond to question 4.

When reviewing the answers, consider having students come to the board or document camera to show the steps of the transformation for question 1 to occur, using a copy of ΔTMW.

5.4.2 Collaboration: Congruence in Terms of Rigid Motions (continued)

2. Complete the table by determining which parts of ΔGXN correspond with the given part of ΔTMW after mapping ΔTMW to ΔGXN.

ΔTMW	ΔGXN
$\overline{MT}$	$\overline{XG}$
$\overline{MW}$	$\overline{XN}$
$\overline{WT}$	$\overline{NG}$
$\angle TMW$	$\angle GXN$
$\angle TWM$	$\angle GNX$
$\angle WTM$	$\angle NGX$

Definition of Congruence in Terms of Rigid Motions

Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

3. Discuss with your partner how to determine if the transformations that were used to map ΔTMW to ΔGXN are rigid motions.
Answers will vary. Yes, two transformations were used, a translation and a reflection. Both of the transformations are rigid motions because they preserve the distance and angle measure.

4. Complete the statement.

ΔTMW and ΔGXN are congruent because there exists

- ○ 1
- ○ 2
- ○ 3

rigid motion(s) which map(s) ΔTMW to ΔGXN.

Consider using colored markers or highlighters to draw attention to the line segments and angles in one triangle and matching the color to its corresponding part in the other triangle. Each part should be represented with a unique color so that students can connect the corresponding parts.

Spend some time reviewing the formal definition of congruence in terms of rigid motions. Don't just read the narration but connect the definition to the work they just did in questions 1 and 2. This formal definition has been informally explored since Unit 1. Address the "if and only if" part of the definition so the students know that both parts must be true. The foundation of this definition was laid in Unit 1, where students explored transformations as the plane moving and how any point on the preimage can be mapped to its corresponding point on the figure.

Why do you think mapping one figure to another shows congruence?

Some students may have seen the sequence of transformations in question 1 as two translations and a reflection. If this is the case, students will respond to question 4 as 3 rigid motions.

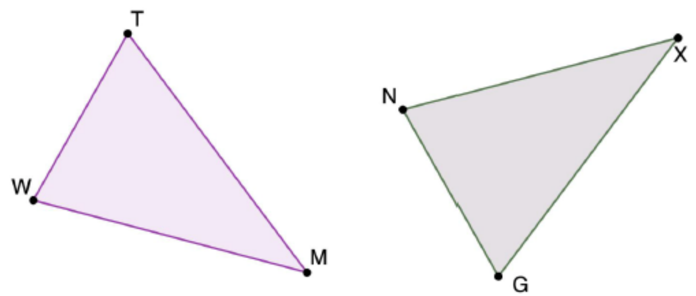
5.4.3 Guided Instruction: CPCTC

Component Overview: *Students will receive guided instruction on Corresponding Parts of Congruent Triangles are Congruent (CPCTC). For this activity, review the context of each problem with students prior to releasing them to try it on their own. Allow students to work with a partner where it's indicated.*

5.4.3 Guided Instruction: CPCTC

Another way to show that triangles are congruent is with congruence postulates and theorems.

1. ΔTMW and ΔGXN are shown where $\angle GNX \cong \angle MWT$, $\angle NXG \cong \angle TMW$, and $\overline{XG} \cong \overline{MT}$.



   a. From the given information, what congruence postulate or theorem can be used to show that $\Delta TMW \cong \Delta GXN$?
   Angle-Angle-Side

   b. Discuss with your partner what could be concluded about the corresponding parts of the triangles if the congruence postulate or theorem is used to show the triangles are congruent.

Corresponding Parts of Congruent Triangles are Congruent (CPCTC)

If two triangles are congruent, then all corresponding parts of the triangles are congruent.

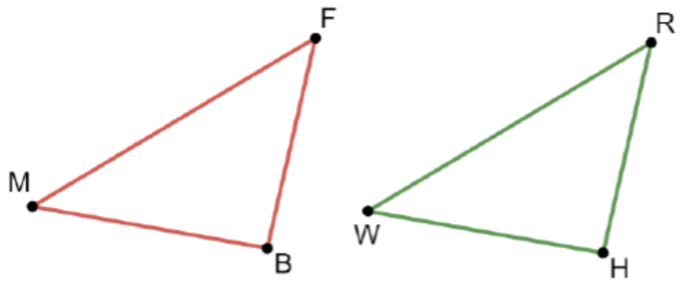
For visual learners, consider modeling how to mark up the triangle using geometric symbols and markings to translate the written information onto the visual of the triangles provided. There are many ways students can mark figures. Encourage students to develop their own strategy that they understand for marking congruent parts.

Consider leveraging a polling protocol where you can poll students on which postulate or theorem can be used to show that the triangles are congruent.
Have students defend their answer or explain their rationale and then do a re-poll to see if anyone was convinced based on the evidence.

5.4.3 Guided Instruction: CPCTC (continued)

There is a difference between naming two triangles and stating that two triangles are congruent. Order is important in a congruence statement, but is not necessary when only naming two triangles. The order used for congruence statements communicates which parts of the triangles correspond.

2. A diagram of two triangles is shown.



Naming	Congruence
ΔMFB and ΔHRW	$\Delta MFB \cong \Delta WRH$

Use the congruence statement to name all of the corresponding parts of the triangles.

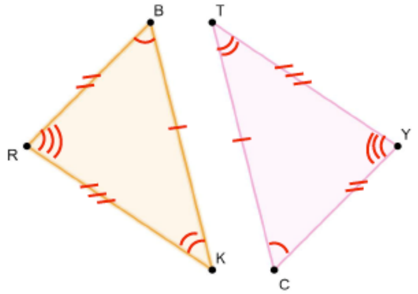
ΔMFB	ΔWRH
$\overline{MF}$	$\overline{WR}$
$\overline{FB}$	$\overline{RH}$
$\overline{MB}$	$\overline{WH}$
$\angle MFB$	$\angle WRH$
$\angle FBM$	$\angle RHW$
$\angle BMF$	$\angle HWR$

When reviewing the definition for Corresponding Parts of Congruent Triangles are Congruent (CPCTC), be sure to emphasize the importance of correctly identifying corresponding parts. Also, connect this concept back to what students did in prior problems.

Is there any way to verify that you are correct?

When reviewing question 2, consider using a document camera and a copy ΔMFB on patty paper to show the corresponding parts.

3. Consider triangles BKR and CTY.



a. If $\Delta BKR \cong \Delta CTY$, mark the corresponding congruent sides and the corresponding congruent angles on the diagram.

b. Complete each congruence statement.

$\overline{BK} \cong$ __$\overline{CT}$___ $\angle KBR \cong \angle$___TCY_

__$\overline{BR}$_____ $\cong \overline{CY}$ $\angle RKB \cong \angle$__YTC__

$\overline{RK} \cong$ __$\overline{YT}$_____ $\angle$__BRK__ $\cong \angle CYT$

4. Consider two congruent triangles, ΔGRD and ΔFHW.

a. Write the congruence statement $\Delta GRD \cong \Delta FHW$ in two other ways.

$\Delta RGD \cong \Delta HFW$ $\Delta DRG \cong \Delta WHF$

b. Name the congruent sides for $\Delta GRD \cong \Delta FHW$.

$\overline{GR} \cong \overline{FH}$ $\overline{RD} \cong \overline{HW}$ $\overline{GD} \cong \overline{FW}$

c. Name the congruent angles for $\Delta GRD \cong \Delta FHW$.

$\angle GRD \cong \angle FHW$

$\angle RDG \cong \angle HWF$

$\angle DGR \cong \angle WFH$

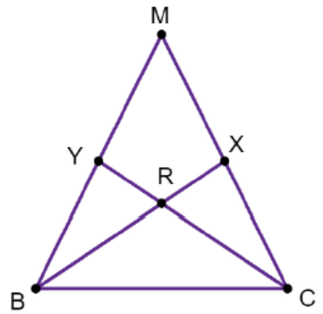
There are many ways that students can mark figures. Two examples are shown (using colors and using tick marks). There is not one "right" way. Rather, what is important is that students have a strategy they understand and use to mark figures after reading given information.
It's helpful to have highlighters on hand if students prefer to use color to mark corresponding parts.

What role does the labeling of the vertices play when determining which parts of a triangle are corresponding parts of the other? How can labeling them incorrectly impact the result?

Why do you think it is possible for triangle congruence statements to be written in multiple ways, yet still be true in terms of congruency?

5.4.3 Guided Instruction: CPCTC (continued)

5. Triangle MBC is shown.



a. How many different triangles are in ΔMBC?
8 triangles

b. If $\Delta BRY \cong \Delta CRX$, then complete the congruence statement.

$\overline{RC} \cong$ __$\overline{RB}$__ $\angle RYB \cong$ _$\angle RXC$_

c. If $\Delta BXM \cong \Delta CYM$, then complete the congruence statement.

$\overline{CY} \cong$ __$\overline{BX}$__ $\overline{CM} \cong$ __$\overline{BM}$__ $\angle MCY \cong$ _$\angle MBX$_

Overlapping triangles can be difficult for some students. Students may not struggle with ΔBRY and ΔCRX, but may struggle with ΔBXM and ΔCYM. Have students draw ΔBRY and ΔCRX separately before marking and then repeat this by having them draw ΔBXM and ΔCYM separately before marking.

When reviewing question 5a, consider using different-colored highlighters to notate each unique triangle in the shape in order for students to better visualize the triangles that exist.

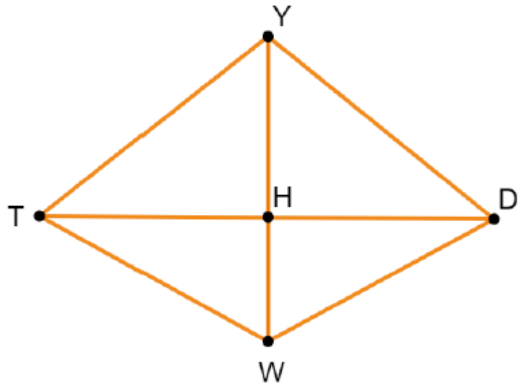
Why shouldn't $\angle RYB$ be simplified to just $\angle Y$ or $\angle RXC$ be simplified to just $\angle X$?

5.4.4 Guided Practice: Using CPCTC

Component Overview: *Students will engage in a practice opportunity on CPCTC. For this activity, review each question and allow students to try to answer independently. Then, review each answer prior to moving forward.*

5.4.4 Guided Practice: Using CPCTC

Quadrilateral $YDWT$ is shown where $\overline{YH}$ is a perpendicular bisector of $\overline{TD}$.



1. How many different triangles are in Quadrilateral $YDWT$?
   8 triangles

2. If $\Delta DHY \cong \Delta THY$, then complete each congruence statement.

   $\overline{HD} \cong$ __$\overline{HT}$__ $\angle DYH \cong$ _$\angle TYH$_ $\overline{TH} \cong$ ___$\overline{DH}$__

3. If $\Delta DHW \cong \Delta THW$, then complete each congruence statement.

   $\angle TWH \cong$ _$\angle DWH$_ $\angle HTW \cong$ _$\angle HDW$_ $\overline{TW} \cong$ __$\overline{DW}$__

4. If $\Delta YTW \cong \Delta YDW$, then complete each congruence statement.

   $\angle WTY \cong$ _$\angle WDY$_ $\overline{DW} \cong$ __$\overline{TW}$__ $\overline{YW} \cong$ __$\overline{YW}$__

Students may struggle with seeing the individual triangles within the given quadrilateral for this activity.

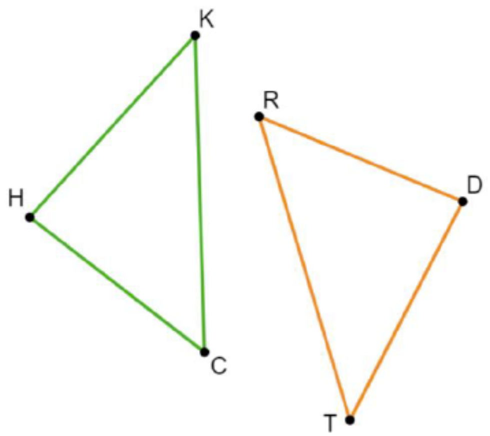
Students who struggle with questions 2-4 may have a misconception about determining corresponding parts of triangles. Review student answers to these questions to see if students may be misinterpreting the way the triangles are written in the congruency statement.

5.4.5 Wrap-Up: Cool Down

Component Overview: *Students will complete a Cool Down that assesses student understanding of the lesson content. This activity should be done independently to determine student mastery.*

5.4.5 Wrap-Up: Cool Down

The diagram shows ΔHKC and ΔTDR where $\Delta HKC \cong \Delta DTR$.



1. Complete each congruence statement.

   $\overline{KC} \cong$ _$\overline{TR}$__ $\overline{CH} \cong$ _$\overline{RD}$_

   $\angle RTD \cong$ _$\angle CKH$ _ $\angle RDT \cong$ __$\angle CHK$ _

2. Becky told her teacher that there was a typo in the first sentence. She circled the typo.

   The diagram shows ΔHKC and ΔTDR where $\Delta HKC \cong \Delta DTR$.

   Write a note to Becky explaining why this is NOT a typo.
   Answers will vary. ΔHKC and ΔTDR is naming the triangles, so order does not matter. The order of the vertices only matters when declaring congruence.

The data from the student response to this activity can provide insight on whether or not the students need additional support on the content from this lesson.
If students miss question 1, they likely misunderstand the connection between the congruence statements and their corresponding sides and angles. Encourage students to mark the congruent parts of the triangles for help in assessing whether students understand the concept of congruent parts but not the congruency statement.
If students miss question 2, they may have forgotten that in the naming of triangles, order does not matter.